Febrile Seizures

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Definitions

Febrile Seizure

Seizure accompanied by fever *without* central nervous system infection, metabolic or electrolyte disturbances, or a history of afebrile seizure or any acute neurological insult/head trauma in children aged 6 months to 6 years. Few guidelines include younger children up to 3 months [National Institutes of Health (NIH)] and even 1 month [International League against Epilepsy (ILAE)] after ruling out causes of provoked seizures. Fever can occur anytime during or *after* a seizure and the majority of febrile seizures (FSs) occur within 24 hours of fever onset.

Simple Febrile Seizures

Fever with isolated, generalized tonic clonic seizures, which last <15 minutes, and do not recur within 24 hours.

Complex Febrile Seizures

Fever with seizures with any one of the following features: focal and/or prolonged for >15 minutes and/or recur within 24 hours and/or have incomplete recovery within 1 hour.

Febrile Status Epilepticus

Febrile seizure lasting for 30 minutes or more and/or series of seizures without full recovery in between that.

Definitions

Afebrile Febrile Seizure

Seizures in an acute infectious illness (particularly gastroenteritis) without documented fever and features consistent with simple FS.

Febrile Seizure Plus

Febrile seizures that continue past the usual age where they are expected to resolve (6 years) and/or accompanied by afebrile generalized or focal seizures.

Genetic Epilepsy with Febrile Seizure Plus

Febrile seizures plus with a family history of FSs, FSs plus, or afebrile generalized or focal seizures.

Pathophysiology of Febrile Seizures

Several hypotheses have been proposed to explain causation of FSs. It is believed that in genetically predisposed children, increase in brain temperature leads to perturbation of temperature sensitive ion channels that in turn causes increased neuronal firing. Moreover, interleukin-1β acts as both a pyrogen and seizure provocator, acting at glutamate pathway. Interleukin-1β is also an NMDA agonist. Last, but not the least, hyperthermia-induced brain alkalosis is also believed to result in neuronal excitability.

Diagnosis of Febrile Seizure

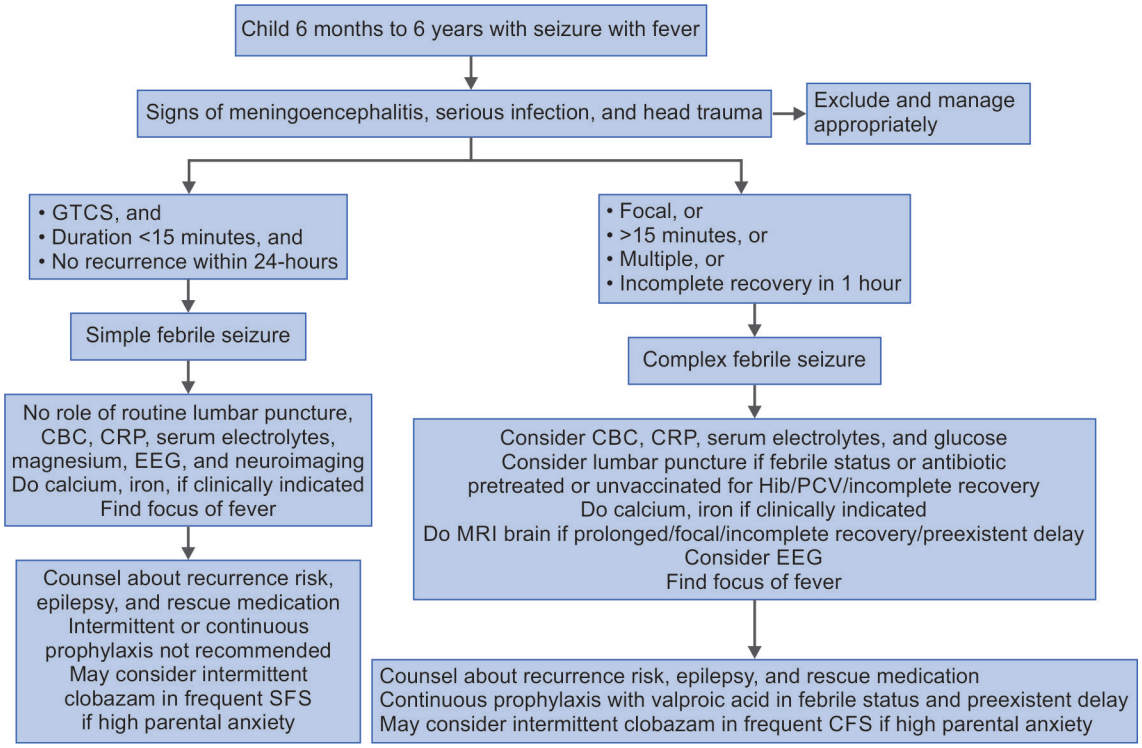
Diagnosis of febrile seizure is depicted in **Table 1**.

TABLE 1: Diagnosis of febrile seizure (FS).

History	*Clinical examination*	*Red flag signs*
☑ Nature and duration of the convulsions and postictal phase ☑ Recent fever/ear discharge/dysuria ☑ Recent antibiotic therapy/ antipyretics/rescue anticonvulsants *Immunization history* ☑ Past history of previous episodes of FS, a diagnosis of epilepsy, and other neurologic conditions and diseases ☑ Family history of FS, epilepsy, or neurologic diseases ☑ History of neonatal intensive care unit stay or developmental delay, if any	☑ Vitals ☑ Obvious focus of infection ☑ Anthropometry ☑ Features of raised intracranial pressure ☑ Features of meningitis like bulging fontanelle, neck retraction, or meningeal signs ☑ Neurocutaneous markers ☑ Dysmorphism ☑ Focal neurological signs ☑ Meningeal signs	☑ Focal neurological signs ☑ Persistent altered sensorium after 1 hour of seizure ☑ Features of raised intracranial pressure such as headache, vomiting, papilledema, brisk deep tendon reflexes, Cushing's triad of bradycardia, irregular respiration, and hypertension ☑ Features of meningoencephalitis/ non-blanching rash in an unwell child ☑ Features of sepsis/shock/ respiratory distress

Evaluation of a Child with First Episode of Febrile Seizure

Flowchart 1: Algorithm for evaluation and management of febrile seizure.



(CBC: complete blood count; CFS: complex febrile seizure; CRP: C-reactive protein; EEG: electro-encephalography; GTCS: generalized tonic-clonic seizure; Hib: *Haemophilus influenzae* type B; PCV: pneumococcal conjugate vaccine)

TABLE 2: Evaluation of a child with first episode febrile seizure.

	Simple febrile seizure (SFS)	***Complex febrile seizure (CFS)/ febrile status epilepticus (FSE)***
Basic history, examination, and red flags **(Table 1)**	Yes	Yes
Lumbar puncture	If features of meningitis; children <12 months of age who have not received the Hib or pneumococcal vaccines, children pretreated with antibiotics or children with severe protein energy malnutrition	Consider in children who remain obtunded after 1 hour of seizure
Neuroimaging	Neuroimaging is routinely not recommended. Consider neuroimaging prior to lumbar puncture in children with focal neurological deficits or clinical symptoms and signs of raised intracranial pressure	MRI brain should be preferably considered within 72 hours for features of viral encephalitis, acute disseminated encephalomyelitis, intracranial space-occupying lesions, cortical malformations, and for hippocampal abnormalities Also do if preexistent developmental delay
Electroencephalography (EEG)	No	Yes, prognostic significance of the abnormalities to predict future epilepsy is unclear. Perform preferably after 48 hours and within 1 week of febrile seizure
Complete blood count with CRP	Not routinely, but may be considered to find cause of fever	Yes
Serum electrolytes	No	Yes
Serum calcium	If <1 year	Yes
Serum magnesium	No	Yes
Blood sugar	No	Yes
Serum iron	If clinical pallor	
Urine analysis	If <18 months or if clinical features suggestive of urinary tract infection (dysuria, frequency, and urgency), if >18 months of age	
Genetic testing for Dravet syndrome (SCN1A, SCN1B, GABRG2, and SCN2A)	Recurrent febrile status epilepticus, onset of prolonged hemiconvulsive seizures below 1 year age, vaccine-associated encephalopathy, or GEFS+	

(CRP: C-reactive protein; GEFS+: genetic epilepsy with febrile seizure plus; Hib: *Haemophilus influenzae* type B)

Acute Management during Seizure

- ☑ The drug of choice for rescue management at home is intranasal midazolam (0.2 mg/kg; maximum: 5 mg). Other effective drugs are intramuscular/buccal midazolam, buccal lorazepam, and per rectal diazepam. Maximum two doses, 5 minutes apart.
- ☑ Management of febrile status epilepticus at hospital is similar to management of convulsive status epilepticus.
- ☑ Stabilize with ABCDE approach (airway, breathing, circulation, disability, and exposure/ examination).
- ☑ If diagnosed with Dravet syndrome, FS+, GEFS+, sodium channel blockers (phenytoin) may be avoided.
- ☑ In young children, in case of clinical suspicion of meningitis and febrile status start third-generation cephalosporin till lumbar puncture results.

Points to Ponder before Institution of Prophylactic Therapy

- ☑ Children with FSs are prone to recurrence.
- ☑ Slightly increased risk of epilepsy in this population is the result of genetic predisposition.
- ☑ No difference in learning has been identified in children with simple FSs, except in those children who had neurologic abnormalities before their first seizure.
- ☑ There is a negligible risk of a child dying during a simple FS due to injury, aspiration, or cardiac arrhythmia.
- ☑ Successful treatment of simple FSs can neither prevent later development of epilepsy nor simple FSs cause any structural damage to the brain.
- ☑ Possible adverse effects of prophylactic therapy include rare fatal hepatotoxicity (especially in children younger than 2 years who are also at greatest risk of FSs), thrombocytopenia, weight loss and gain, gastrointestinal disturbances, and pancreatitis with valproic acid and hyperactivity, irritability, lethargy, sleep disturbances, and hypersensitivity reactions with phenobarbital; lethargy, drowsiness, and ataxia for intermittent diazepam as well as the risk of masking an evolving central nervous system infection.
- ☑ Antipyretics and tepid sponging, if properly done, may improve the comfort of the child, they do not prevent fever or FSs, e.g., paracetamol 15 mg/kg/dose SOS up to 6 hourly.

Prophylactic Regimens

- ☑ Neither continuous nor intermittent anticonvulsant therapy is recommended for children with one or more simple FSs as there is preponderance of harm over benefit with therapy.
- ☑ Intermittent prophylaxis with oral benzodiazepine may be considered among children with frequent recurrent simple FSs with parental anxiety and residence far from medical facilities or complex FS who have not been started on continuous prophylaxis.
- ☑ Continuous prophylaxis with anticonvulsants may be considered among children with febrile status epilepticus, FSs in children with neurodevelopmental delay like cerebral palsy, global developmental delay or autism spectrum disorder, frequent complex FSs, and children with FS+/GEFS+ with afebrile seizures.

Prophylactic Drugs

The drug of choice for intermittent prophylaxis is clobazam (0.5–1 mg/kg/day in two divided doses for 3 days without tapering; maximum dose 20 mg/day).

Drug of choice for continuous prophylaxis is sodium valproate (20–40 mg/kg/day), others are phenobarbital (3–5 mg/kg/day); primidone (15–20 mg/kg/day). Carbamazepine and phenytoin are ineffective.

Duration

The anticonvulsants should be considered for 2 years seizure freedom period or guided individually based on primary syndrome (GEFS+/Dravet syndrome).

TABLE 3: Future prognosis.

Future risk of epilepsy	***Recurrence of febrile seizures***
Children with simple febrile seizures have *1% risk* of developing epilepsy *by the age of 7 years* (same as the general population) and *2.4% by 25 years of age* Children who have had multiple simple febrile seizures are younger than 12 months at the time of their first febrile seizure and have a family history of epilepsy are at higher risk Risk of epilepsy after complex febrile seizures depends on number of complex features: ☑ With 1 complex feature: 6–8% ☑ With 2 complex features: 17–22% ☑ With 3 and more features: 49% Risk factors for developing subsequent epilepsy include: family history of epilepsy, any neuro-developmental problem, prolonged or focal febrile seizures, and febrile status epilepticus	32% children have recurrent seizures 17% have one recurrence 9% have two recurrences 6% have three or more recurrences Recurrence rate within 1 year and 2-year of first febrile seizure is 75% and 90%, respectively Risk factors for recurrent febrile seizures ☑ Early onset <18 months of age ☑ Family history of febrile seizures or epilepsy in a first-degree relative ☑ Low grade fever associated with seizure onset (<39°C) ☑ Short duration of fever before the seizure (<1 hour) ☑ Complex febrile seizure ☑ Attendance at a day care nursery—presumed increased viral exposure

Family History of Febrile Seizure

- ☑ One-third of children with FS have a positive family history.
- ☑ Prevalence risk increase to one in five if one sibling affected and one in three if both parents and sibling have been affected.
- ☑ Concordance rate of FS are higher in monozygotic (35–69%) than dizygotic twins (14–20%).

Further Reading

- ☑ Clinical Practice Guidelines. Subcommittee on Febrile Seizures American Academy of Pediatrics. Neurodiagnostic evaluation of the child with a simple febrile seizure. Pediatrics. 2011;127(2):389-94.
- ☑ Kaushik JS, Sondhi V, Yoganathan S, Dubey R, Sharma S, Vinayan KP, Association of Child Neurology (AOCN) Consensus Statement on the Diagnosis and Management of Febrile Seizures. Indian Pediatr. 2022;59:300-6.
- ☑ Steering Committee on Quality Improvement and Management, Subcommittee on Febrile Seizures American Academy of Pediatrics. Febrile seizures: clinical practice guideline for the long-term management of the child with simple febrile seizures. Pediatrics. 2008;121(6):1281-6.
- ☑ Whelan H, Harmelink M, Chou E, Sallowm D, Khan N, Patil R, et al. Complex febrile seizures—a systematic review. Dis Mon. 2017;63(1):5-23.

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